Student Performance Analysis System using NumPy

Objective

Analyze student marks using NumPy by performing student-wise analysis, subject-wise analysis, class statistics, ranking, filtering, searching, broadcasting, and array manipulation.

In [1]:
```
import numpy as np
```

In [2]:
```
# Array for Student names created
student_names = np.array(["Rahul", "Amit", "Priya", "Mohit", "Aakash","Nisha", "Riy
```

In [3]:
```
# Array for Subject names created
subject_names = np.array(["Maths","Science","English","History","Geography"])
```

In [4]:
```
#Array for Student marks created
marks = np.random.randint(35,101,size=(10,5))
```

Display Complete Dataset

In [5]:
```
print("Name\t", end="")
for subject in subject_names:
    print(subject, end="\t")
print()

for i in range(len(student_names)):
    print(student_names[i], end="\t")
    for mark in marks[i]:
        print(mark, end="\t")
    print()
```

```
Name    Maths   Science English History Geography
Rahul   99      59      94      55      69
Amit    67      89      38      78      40
Priya   39      86      36      43      58
Mohit   89      79      88      76      56
Aakash  90      41      79      55      64
Nisha   65      47      99      47      62
Riya    61      45      41      48      57
Sahil   81      89      61      79      57
Aditya  83      75      37      62      46
Kavya   84      42      89      76      53
```

Dataset Information

In [6]:
```
print("Shape :", marks.shape)
print("Dimensions :", marks.ndim)
print("Total Elements :", marks.size)
print("Data Type :", marks.dtype)
print("Memory Used :", marks.size * marks.itemsize, "Bytes")
print("Object Type :", type(marks))
```

```
Shape : (10, 5)
Dimensions : 2
Total Elements : 50
Data Type : int32
Memory Used : 200 Bytes
Object Type : <class 'numpy.ndarray'>
```

Student Wise Analysis

In [7]:
```
for i in range(len(student_names)):
    print("*"*35)
    print("Student Name :", student_names[i])
    print()

    for j in range(len(subject_names)):
        print(f"{subject_names[j]} : {marks[i,j]}")

    print()
    print("Total Marks :", np.sum(marks[i]))
    print("Average :", np.mean(marks[i]))
    print("Highest :", np.max(marks[i]))
    print("Lowest :", np.min(marks[i]))
    print("Percentage :", np.sum(marks[i])/500*100,"%")
    print()
```

```
***********************************
Student Name : Rahul

Maths : 99
Science : 59
English : 94
History : 55
Geography : 69

Total Marks : 376
Average : 75.2
Highest : 99
Lowest : 55
Percentage : 75.2 %

***********************************
Student Name : Amit

Maths : 67
Science : 89
English : 38
History : 78
Geography : 40

Total Marks : 312
Average : 62.4
Highest : 89
Lowest : 38
Percentage : 62.4 %

***********************************
Student Name : Priya

Maths : 39
Science : 86
English : 36
History : 43
Geography : 58

Total Marks : 262
Average : 52.4
Highest : 86
Lowest : 36
Percentage : 52.400000000000006 %

***********************************
Student Name : Mohit

Maths : 89
Science : 79
English : 88
History : 76
Geography : 56

Total Marks : 388
Average : 77.6
```

```
Highest : 89
Lowest : 56
Percentage : 77.60000000000001 %

**********************************
Student Name : Aakash

Maths : 90
Science : 41
English : 79
History : 55
Geography : 64

Total Marks : 329
Average : 65.8
Highest : 90
Lowest : 41
Percentage : 65.8 %

**********************************
Student Name : Nisha

Maths : 65
Science : 47
English : 99
History : 47
Geography : 62

Total Marks : 320
Average : 64.0
Highest : 99
Lowest : 47
Percentage : 64.0 %

**********************************
Student Name : Riya

Maths : 61
Science : 45
English : 41
History : 48
Geography : 57

Total Marks : 252
Average : 50.4
Highest : 61
Lowest : 41
Percentage : 50.4 %

**********************************
Student Name : Sahil

Maths : 81
Science : 89
English : 61
History : 79
```

```
Geography : 57

Total Marks : 367
Average : 73.4
Highest : 89
Lowest : 57
Percentage : 73.4 %

***********************************
Student Name : Aditya

Maths : 83
Science : 75
English : 37
History : 62
Geography : 46

Total Marks : 303
Average : 60.6
Highest : 83
Lowest : 37
Percentage : 60.6 %

***********************************
Student Name : Kavya

Maths : 84
Science : 42
English : 89
History : 76
Geography : 53

Total Marks : 344
Average : 68.8
Highest : 89
Lowest : 42
Percentage : 68.8 %
```

Subject Wise Analysis

In [8]:

```
for i in range(len(subject_names)):
    print("*"*35)
    print(subject_names[i])
    print()

    for j in range(len(student_names)):
        print(f"{student_names[j]}\t{marks[j,i]}")

    print()
    print("Highest :", np.max(marks[:,i]))
    print("Lowest :", np.min(marks[:,i]))
    print("Average :", np.mean(marks[:,i]))
```

```
    print("Total :", np.sum(marks[:,i]))
    print()
```

```
**********************************
Maths

Rahul   99
Amit    67
Priya   39
Mohit   89
Aakash  90
Nisha   65
Riya    61
Sahil   81
Aditya  83
Kavya   84

Highest : 99
Lowest : 39
Average : 75.8
Total : 758

**********************************
Science

Rahul   59
Amit    89
Priya   86
Mohit   79
Aakash  41
Nisha   47
Riya    45
Sahil   89
Aditya  75
Kavya   42

Highest : 89
Lowest : 41
Average : 65.2
Total : 652

**********************************
English

Rahul   94
Amit    38
Priya   36
Mohit   88
Aakash  79
Nisha   99
Riya    41
Sahil   61
Aditya  37
Kavya   89

Highest : 99
Lowest : 36
Average : 66.2
Total : 662
```

```
***********************************
History

Rahul   55
Amit    78
Priya   43
Mohit   76
Aakash  55
Nisha   47
Riya    48
Sahil   79
Aditya  62
Kavya   76

Highest : 79
Lowest : 43
Average : 61.9
Total : 619

***********************************
Geography

Rahul   69
Amit    40
Priya   58
Mohit   56
Aakash  64
Nisha   62
Riya    57
Sahil   57
Aditya  46
Kavya   53

Highest : 69
Lowest : 40
Average : 56.2
Total : 562
```

Class Report

In [9]:

```
print("*"*35)
print("CLASS REPORT")
print("*"*35)

print("Total Students :", len(student_names))
print("Total Subjects :", len(subject_names))
print()

print("Total Marks :", np.sum(marks))
print("Average :", np.mean(marks))
print("Median :", np.median(marks))
print("Variance :", np.var(marks))
print("Standard Deviation :", np.std(marks))
```

```
print()

print("Highest Marks :", np.max(marks))
print("Lowest Marks :", np.min(marks))
```

```
***********************************
CLASS REPORT
***********************************
Total Students : 10
Total Subjects : 5

Total Marks : 3253
Average : 65.06
Median : 62.0
Variance : 346.37639999999993
Standard Deviation : 18.611190182253253

Highest Marks : 99
Lowest Marks : 36
```

Topper

In [10]:

```
student_totals = np.sum(marks, axis=1)

print("Topper :", student_names[np.argmax(student_totals)])
print("Total :", np.max(student_totals))
print("Percentage :", np.max(student_totals)/500*100,"%")
```

```
Topper : Mohit
Total : 388
Percentage : 77.60000000000001 %
```

Ranking

In [11]:

```
student_totals = np.sum(marks, axis=1)

rank = np.argsort(student_totals)

print("*"*10,"CLASS RANK","*"*10)

r = 1
for i in rank[::-1]:
    print(f"{r}. {student_names[i]}\t{student_totals[i]}")
    r += 1
```

```
********** CLASS RANK **********
1. Mohit        388
2. Rahul        376
3. Sahil        367
4. Kavya        344
5. Aakash       329
6. Nisha        320
7. Amit 312
8. Aditya       303
9. Priya        262
10. Riya        252
```

Boolean Indexing

In [12]:
```
print("Students scoring above 350")
print(student_names[student_totals > 350])
print(student_totals[student_totals > 350])
```

```
Students scoring above 350
['Rahul' 'Mohit' 'Sahil']
[376 388 367]
```

Searching using np.where()

In [13]:
```
print("100 in Maths")
print(student_names[np.where(marks[:,0] == 100)])

print("\nLess than 40 in English")
print(student_names[np.where(marks[:,2] < 40)])

print("\nExactly 50 in Science")
print(student_names[np.where(marks[:,1] == 50)])

print("\n90 or above in Geography")
print(student_names[np.where(marks[:,4] >= 90)])
```

```
100 in Maths
[]

Less than 40 in English
['Amit' 'Priya' 'Aditya']

Exactly 50 in Science
[]

90 or above in Geography
[]
```

Broadcasting

In [14]:
```
grace_marks = marks + 5
grace_marks = np.where(grace_marks > 100, 100, grace_marks)
```

```
print("Grace Marks")
print(grace_marks)

geo_grace = marks[:,4] + 10
geo_grace = np.where(geo_grace > 100, 100, geo_grace)

print("\nGeography Grace Marks")
print(geo_grace)
```

```
Grace Marks
[[100  64  99  60  74]
 [ 72  94  43  83  45]
 [ 44  91  41  48  63]
 [ 94  84  93  81  61]
 [ 95  46  84  60  69]
 [ 70  52 100  52  67]
 [ 66  50  46  53  62]
 [ 86  94  66  84  62]
 [ 88  80  42  67  51]
 [ 89  47  94  81  58]]

Geography Grace Marks
[79 50 68 66 74 72 67 67 56 63]
```

Append

In [15]:
```
student_names = np.append(student_names,"Anjali")

new_marks = np.array([91,88,95,90,87]).reshape(1,5)
marks = np.append(marks,new_marks,axis=0)

print(student_names)
print(marks.shape)
```

```
['Rahul' 'Amit' 'Priya' 'Mohit' 'Aakash' 'Nisha' 'Riya' 'Sahil' 'Aditya'
 'Kavya' 'Anjali']
(11, 5)
```

Insert

In [16]:
```
student_names = np.insert(student_names,5,"Rohit",axis=0)
marks = np.insert(marks,5,[91,88,95,90,87],axis=0)

print(student_names)
print(marks.shape)
```

```
['Rahul' 'Amit' 'Priya' 'Mohit' 'Aakash' 'Rohit' 'Nisha' 'Riya' 'Sahil'
 'Aditya' 'Kavya' 'Anjali']
(12, 5)
```

Delete

In [17]:
```
subject_names = np.delete(subject_names, subject_names=="History")
marks = np.delete(marks,3,axis=1)

print(subject_names)
print(marks.shape)
```

```
['Maths' 'Science' 'English' 'Geography']
(12, 4)
```

Concatenate

In [18]:
```
a = np.array([1,2,3])
b = np.array([4,5,6])

print(np.concatenate((a,b)))
```

```
[1 2 3 4 5 6]
```